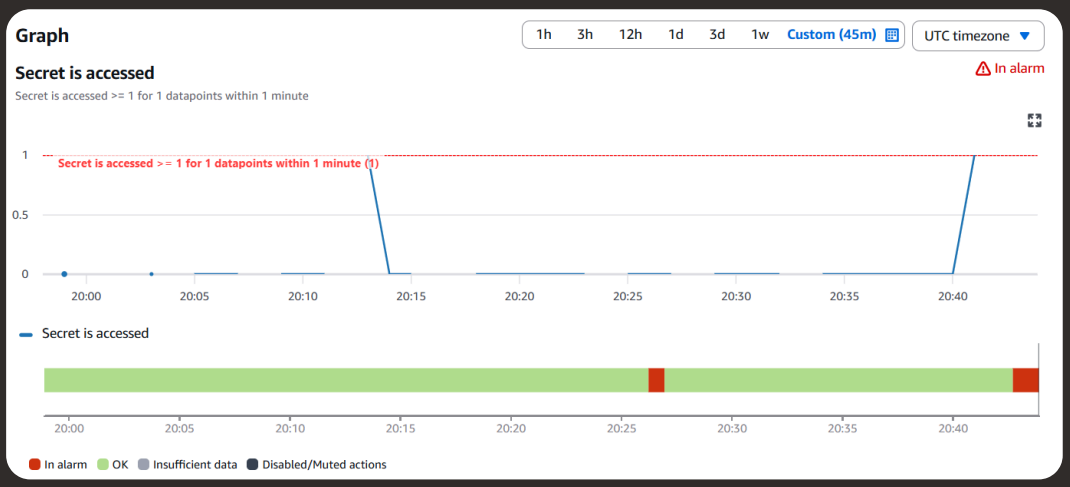
BUILD A SECURITY MONITORING SYSTEM

By Hayder Ben Abdelhamid | October 2026



Introducing Today's Project!

In this project, I will demonstrate how to set up a monitoring system in AWS using CloudTrail, CloudWatch and SNS! I'm doing this project to learn how security and monitoring services in AWS work, plus have a working system thta actually sed us an email.

Tools and concepts

Services I used were CloudTrail, CloudWatch and SNS. I also used Secrets Manager, IAM (roles) and S3 buckets. Key concepts I learnt include secret storing, CloudWatch vs CloudTrail, what are notification and different kinds of endpoints.

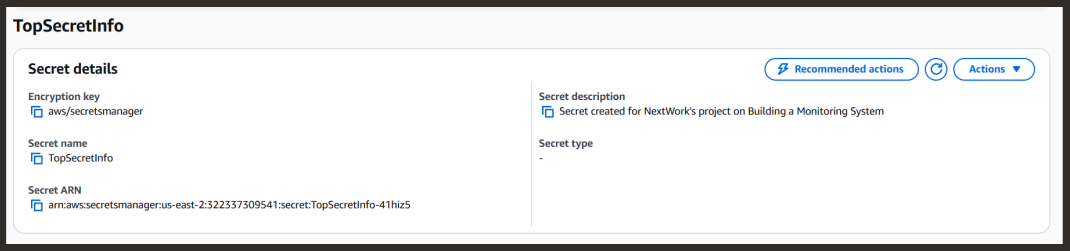
Project reflection

This project took me approximately 3 Hours. The most challenging part was to troubleshooting why the email wasn't delivering in our first test, it would be frustruating when an error is happening but there are no 'error logs' or error messages. It was most rewarding to compare CloudTrail, SNS notifications it really opens up your mind around why we had to use CloudWatch and alarms too.

Create a Secret

Secrets Manager is AWS security services for storing secrets like databases, creadentials, account IDs, API keys...
anything that is sensetives information that would cause any damage trouble if it got leaked ans shoudn't be lying around in code.

To set up for my project, I created a secret called TopSecretInfo in Secrets Manager. This secret is a string that contairs a hot take from us . i already mention thati i need 3 coffees a dy to function.



Set Up CloudTrail

CloudTrail is a monitoring service, it's used to track events and activities in my AWS account. These longs are very helpful for security (i.e. detecting suspicious activity), compliance (i.e. proving that you're following the rules for somethings), and troubleshooting (i.e. identifiying what happened/changed if something breaks).

CloudTrail events include types like management, data, insights and network activity events. In this project, i set up my trail Management events because accessing a secret falls into that catefory. It is not a data event (which captures high volume actions perfomed on resources) because all management events are free to track (and AWS lets us track security operations like this for free).

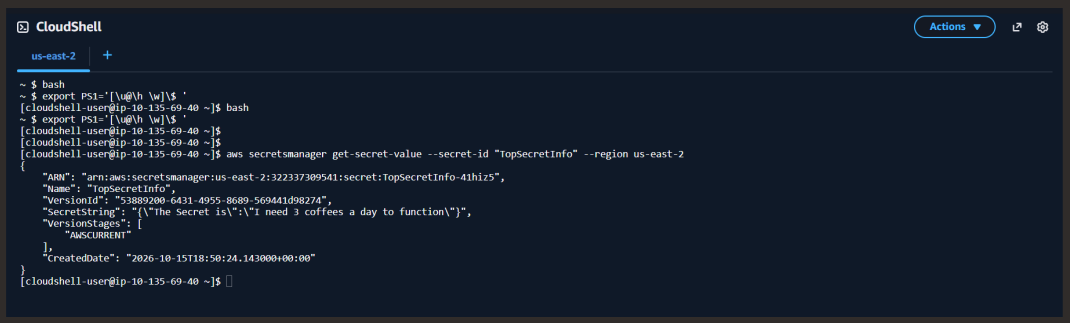
Read vs Write Activity

Read API activity involves accessing, reading, opening a resource. Write API activity involves creating, deleting, updating a resource. For this project, i ticked both to leran about both types of activities, but i really only need the Write activity (accessing a secret is considered a Write activity because ot its importance)

Verifying CloudTrail

I retrieved the secret in two ways: First through the Secrets Manager console, where i could easily just select a 'Retrleve secrets value' button. Second using the AWS CLI i.e. running a get-secret-value in CloudShell.

To analyze my CloudTrail events i.e. see the event where we get the secret's value, i visited the Event history in CloudTrail. I found that there was a GetSecretValue event tracked regardless of whether i did it over the CLI or over the console. This tells us that CloudTrail can definitly see and track when we open the Secrets Manger key

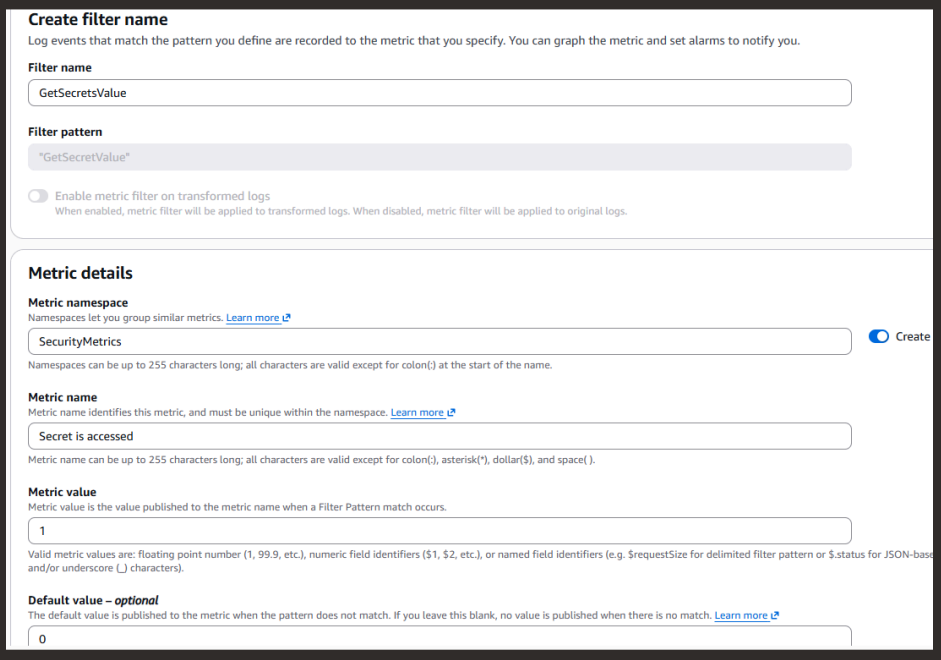


CloudWatch Metrics

CloudWatch Logs is a monitoring service that brings together logs from other AWS services (including CloudTrail) to help us analyze and create alarms for. It's important for monitoring because you get to create insights and get alerted based on events that happen in your account.

CloudTrail's Event History is useful for quickly reading (management) events that happened in the last 90 days, while CloudWatch Logs are better for combining and analysing logs from different sources, accessing logs for longer than 90 days and advaced filtering.

A CloudWatch metric is a specific way that we count or track events that are in a log group. When setting up a metric, the metric value represents how we increment or 'count' an event when it passes our filters (in this case, i wanted to incriment the metric value by 1 whenever our secret is accessed).
Default value is used when the event that we're tracking does not occur.

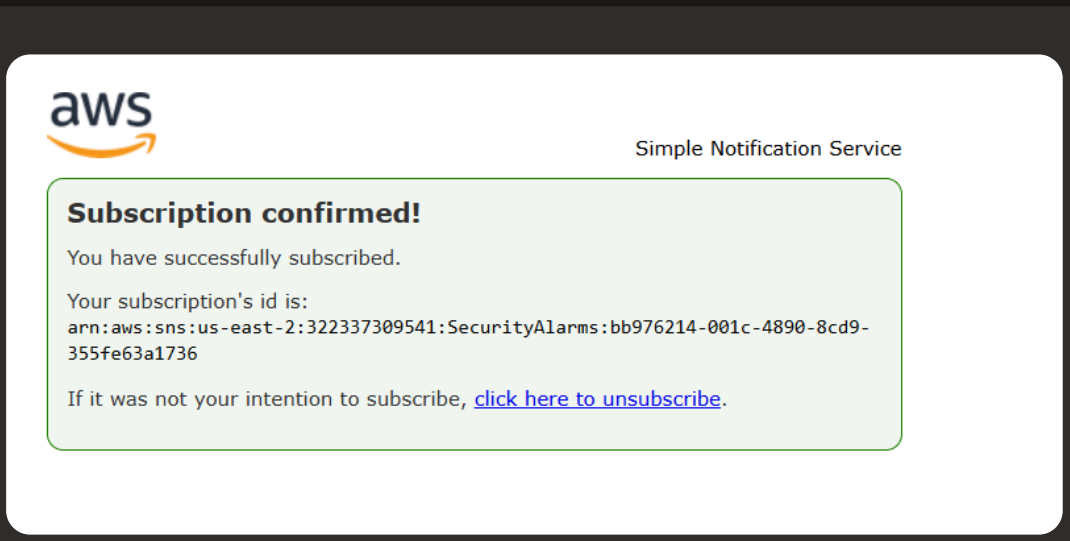


CloudWatch Alarm

A CloudWatch alarm is a treature and alert system in CloudWatch that'q designed to "go off" indicate when certain conditions have been met in our log group. I set my CloudWatch alarm threshold to be about how many times the GetSecretValue event happens in a 5 minute period so the alarm will trigger when the average number of times is above one.

I created an SNS topic along the way. An SNS topic is like a newsletter/broadcast channel thet emails, phone numbers, functions. apps can subscribe to (so thet get notified when SNS has a new update to share).
My SNS topic is set up to send us an email when my secret gets accessed.

AWS requires email confirmation because it would not automatically start emailling addresses that we subscribes to an SNS topic. This helps prevent any unwanted subscriptions for recipicients i.e. people who are receiving those emails.



Troubleshooting Notification Errors

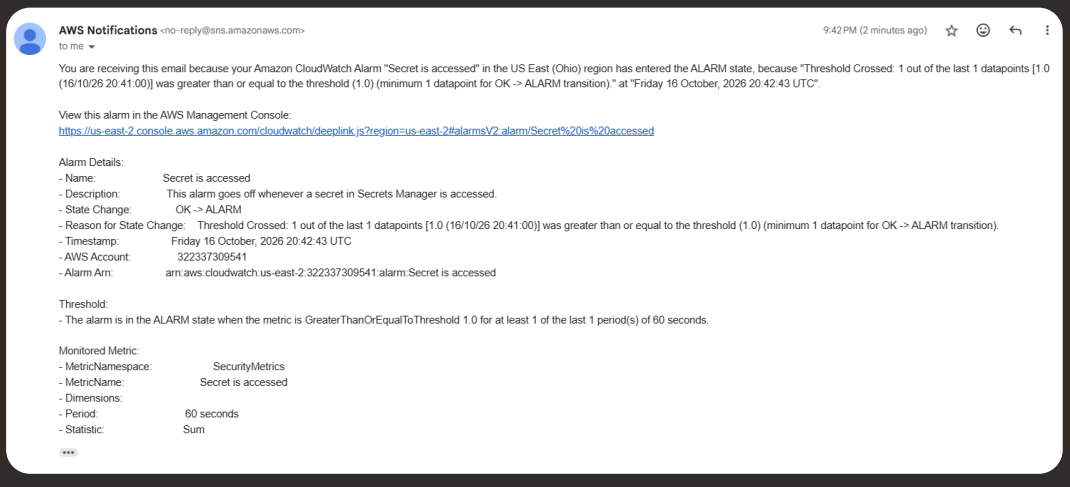
To test my monitoring system, I opened and accessed my secret again! The results weren't successful i didn't get any emails/notifications about the secret getting accessed.

When troubleshooting the notification issues I investigated every single part of our monitoring system whether CloudTrail is picking up on events that are happening when i access mu secret, whether CloudTrail is sending logs CloudWatch, whether the filter is accidentally rejecting the correct events, whether the alarm gets triggered, whether the triggering the alarm sends an email.

I initially didn't receive an email before because CloudWatch was configured to use the wrong threshold instead of calculating the AVERAGE number of times a secret was accessed in a time period, it should've been the SUM!

Success!

To validate that my monitoring system can successfully detect and alert when my secret is accessed.
I checked my secret's value one more time. I received an email within 1-2 minutes of the event! my alarm in CloudWatch is also in an alarm state.



Comparing CloudWatch with CloudTrail Notifications

In a project extension, I enabled SNS notification delivery in CloudTrail because this lets us evaluate CloudTrail vs CloudWatch for notifiying about events like our secret getting acces

After enabling CloudTrail SNS notifications, my inbox was very quickly filled with new emails from SNS (as it was notified by CloudTrail). In terms of the usefulness of these emails, I thought that i'm receiving (it's a littel overwherlming) and the logs themselves don't show what happened in terms of management events that ocurred. I only see that new logs have been stored in our bucket.

